Bicol University College of Science
Department of Information Technology

IT106 - Application Development in Emerging Technologies
2nd Semester 2025-2026

Activity:

Group REST API Enhancement and Multi-Client Implementation

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GROUP DEMONSTRATION

I. Console Application Client

1.1 Teacher-side Pages

```
Student Management Python Client

  Shared Portal
  1. Register
  2. Login
  3. Exit
  Choose an option:
```

Figure 1.1.1 Shared Menu

```
Student Management Python Client

Shared Portal
1. Register
2. Login
3. Exit
Choose an option: 1

Choose account role
1. Teacher
2. Student
Choose an option: 1
Account ID: ABC123
Full name: Charles Babbage
Password: analytical_Engine01

Account registered successfully.

Press Enter to continue...
```

Figure 1.1.2 Registering a Teacher

```
Shared Portal
1. Register
2. Login
3. Exit
Choose an option: 1

Choose account role
1. Teacher
2. Student
Choose an option: 1
Account ID: ABC123

Account ID already exists.
```

Figure 1.1.3 Registering the same teacher account ID (API returns duplicate registration response)

```
Shared Portal
1. Register
2. Login
3. Exit
Choose an option: 2
Account ID: ABC124
Password: haha

Account not found.

Press Enter to continue...
```

Figure 1.1.4 Logging in with invalid account ID (API returns not found response)

```
Shared Portal
1. Register
2. Login
3. Exit
Choose an option: 2
Account ID: ABC123
Password: analytical_Engine02

Invalid account ID or password.

Press Enter to continue...
```

Figure 1.1.5 Logging in with incorrect password (API returns invalid credentials response)

```
Shared Portal
1. Register
2. Login
3. Exit
Choose an option: 2
Account ID: ABC123
Password: analytical_Engine01

Login successful.

Teacher Dashboard - Charles Babbage (ABC123)
1. View my courses
2. Create a course
3. Enter a course
4. Logout
Choose an option:
```

Figure 1.1.6 Teacher Dashboard after Login

```
Teacher Dashboard - Charles Babbage (ABC123)
1. View my courses
2. Create a course
3. Enter a course
4. Logout
Choose an option: 1

Courses retrieved successfully.

No courses created yet.

Press Enter to continue...
```

Figure 1.1.7 Viewing Courses (No Courses Found API response)

```
Teacher Dashboard - Charles Babbage (ABC123)
1. View my courses
2. Create a course
3. Enter a course
4. Logout
Choose an option: 2
Enter the course code: IT-101
Enter the course name: Computer Programming 1

Course created successfully.

Details:
 - Course ID: 5
 - Course Code: IT-101
 - Course Name: Computer Programming 1
 - Teacher: Charles Babbage (ABC123)
 - Course Key: DB3B7V
 - Registered Students: 0
 - Created At: 2026-04-17 11:27:25

Press Enter to continue...
```

Figure 1.1.8 Creating a course (random course key is generated)

```
Teacher Dashboard - Charles Babbage (ABC123)
1. View my courses
2. Create a course
3. Enter a course
4. Logout
Choose an option: 1

Courses retrieved successfully.

Courses:
 - [5] IT-101 - Computer Programming 1 | Key: DB3B7V | Students: 0
```

Figure 1.1.9 Successful viewing of courses

```
Teacher Dashboard - Charles Babbage (ABC123)
1. View my courses
2. Create a course
3. Enter a course
4. Logout
Choose an option: 3

Courses retrieved successfully.

Courses:
 - [5] IT-101 - Computer Programming 1 | Key: DB3B7V | Students: 0

Enter the course ID to open: IT-101

Teacher Course Menu
1. View course details
2. View all students
3. Search or verify a student
4. Delete a student from this course
5. Back
Choose an option:
```

Figure 1.1.10 Enter a Course / Specific Course Dashboard

```
Teacher Course Menu
1. View course details
2. View all students
3. Search or verify a student
4. Delete a student from this course
5. Back
Choose an option: 1

Course details retrieved successfully.

Details:
 - Course ID: 5
 - Course Code: IT-101
 - Course Name: Computer Programming 1
 - Teacher: Charles Babbage (ABC123)
 - Course Key: DB3B7V
 - Registered Students: 0
 - Created At: 2026-04-17 11:27:25

Press Enter to continue...
```

Figure 1.1.11 Viewing a Course Details

```
Teacher Course Menu
1. View course details
2. View all students
3. Search or verify a student
4. Delete a student from this course
5. Back
Choose an option: 2

Students retrieved successfully.

Details:
 - Course ID: 5
 - Course Code: IT-101
 - Course Name: Computer Programming 1
 - Teacher: Charles Babbage (ABC123)
 - Course Key: DB3B7V
 - Registered Students: 0
 - Created At: 2026-04-17 11:27:25

No students are enrolled in this course yet.

Press Enter to continue...
```

Figure 1.1.12 Viewing Course Students (No student enrolled)

```
Teacher Course Menu
1. View course details
2. View all students
3. Search or verify a student
4. Delete a student from this course
5. Back
Choose an option: 3
Enter the student ID to search: dffd

Student lookup complete.

Result:
 - Student not enrolled in the course.

Press Enter to continue...
```

Figure 1.1.13 Student not Enrolled API Response

```
Teacher Course Menu
1. View course details
2. View all students
3. Search or verify a student
4. Delete a student from this course
5. Back
Choose an option: 4
Enter the student ID to remove: ffdf

Student account not found.

Press Enter to continue...
```

Figure 1.1.14 Deleting a Student from Course (Student not Found)

1.2 Student-side Functionalities

```
Shared Portal
1. Register
2. Login
3. Exit
Choose an option: 1

Choose account role
1. Teacher
2. Student
Choose an option: 2
Account ID: ABC124
Full name: Ada Lovelace
Password: bernoulli_Algorithm01

Account registered successfully.

Press Enter to continue...
```

Figure 1.2.1 Registering a Student (same validations as teacher registration)

```
Shared Portal
1. Register
2. Login
3. Exit
Choose an option: 2
Account ID: ABC124
Password: bernoulli_Algorithm01

Login successful.
```

Figure 1.2.2 Logging in as a Student / Student Dashboard (same validations as teacher login)

```
Student Dashboard - Ada Lovelace (ABC124)
1. View enrolled courses
2. Enroll to a course
3. Enter a course
4. Unenroll from a course
5. Logout
```

Figure 1.2.3 Student Dashboard (All options contain same error response when no data is available)

```
Student Dashboard - Ada Lovelace (ABC124)
1. View enrolled courses
2. Enroll to a course
3. Enter a course
4. Unenroll from a course
5. Logout
Choose an option: 2
Enter the course key/password: DB3B7V

Student enrolled successfully.

Details:
 - Course ID: 5
 - Course Code: IT-101
 - Course Name: Computer Programming 1
 - Teacher: Charles Babbage (ABC123)
 - Course Key: DB3B7V
 - Students In Course: 1
 - Created At: 2026-04-17 11:27:25

Press Enter to continue...
```

Figure 1.2.4 Enrolling a student to a course (course key is required)

```
Student Dashboard - Ada Lovelace (ABC124)
1. View enrolled courses
2. Enroll to a course
3. Enter a course
4. Unenroll from a course
5. Logout
Choose an option: 3

Courses retrieved successfully.

Courses:
 - [5] IT-101 - Computer Programming 1 | Teacher: Charles Babbage | Key: DB3B7V

Enter the course ID number or course code to open: IT-101

Student Course Menu
1. View course details
2. View all classmates
3. Search or verify a classmate
4. Unenroll from this course
5. Back
Choose an option:
```

Figure 1.2.5 Entering a Course (Student-side course Dashboard)

```
2. View all classmates
3. Search or verify a classmate
4. Unenroll from this course
5. Back
Choose an option: 2

Classmates retrieved successfully.

Details:
 - Course ID: 5
 - Course Code: IT-101
 - Course Name: Computer Programming 1
 - Teacher: Charles Babbage (ABC123)
 - Course Key: DB3B7V
 - Students In Course: 4
 - Created At: 2026-04-17 11:27:25

Classmates:
 - ABC127 - Alan_Turing
 - ABC125 - George Boole
 - ABC126 - Tim Berners-Lee

Press Enter to continue...
```

Figure 1.2.6 Viewing all Classmates in a course (students were added)

```
Student Dashboard - Ada Lovelace (ABC124)
1. View enrolled courses
2. Enroll to a course
3. Enter a course
4. Unenroll from a course
5. Logout
Choose an option: 4

Courses retrieved successfully.

Courses:
 - [5] IT-101 - Computer Programming 1 | Teacher: Charles Babbage | Key: DB3B7V

Enter the course ID number or course code to unenroll from: IT-101

Student unenrolled from the course successfully.

Result:
 - Course ID: 5
 - Course Code: IT-101
 - Course Name: Computer Programming 1

Press Enter to continue...
```

Figure 1.2.7 Unenrolling From a Course

II. Web Application Client

2.1 Teacher-side Functionalities

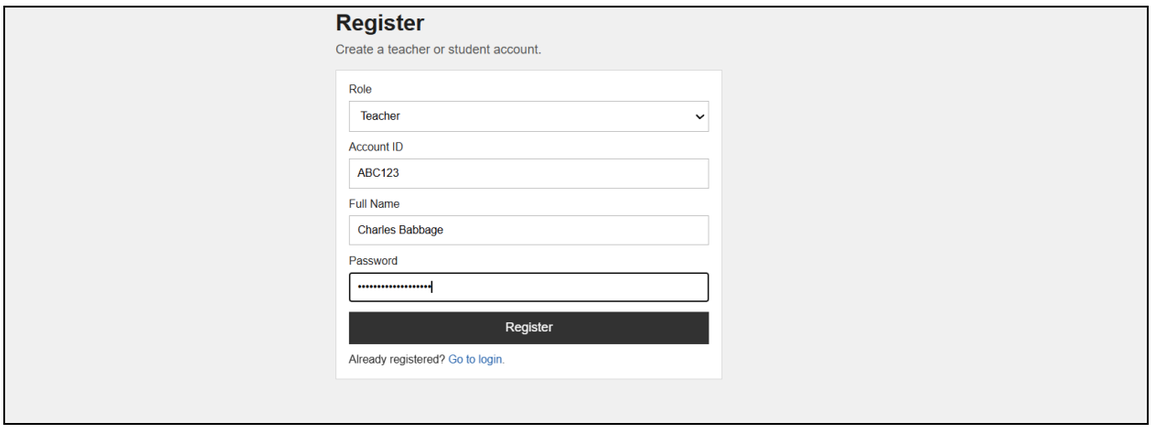


Figure 2.1.1 Teacher Registration

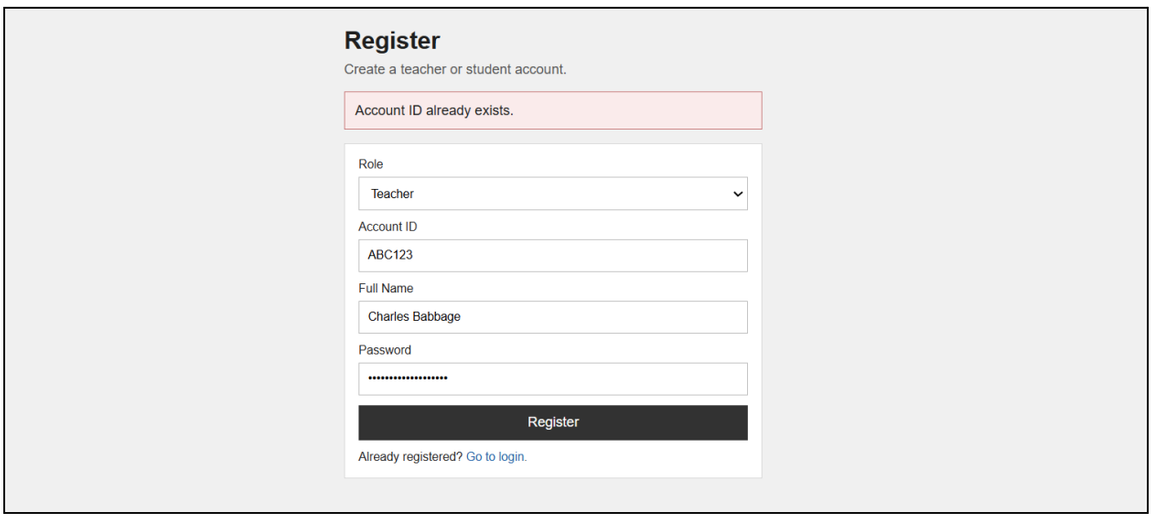


Figure 2.1.2 Registering the Same Account (API returns duplicate account response)

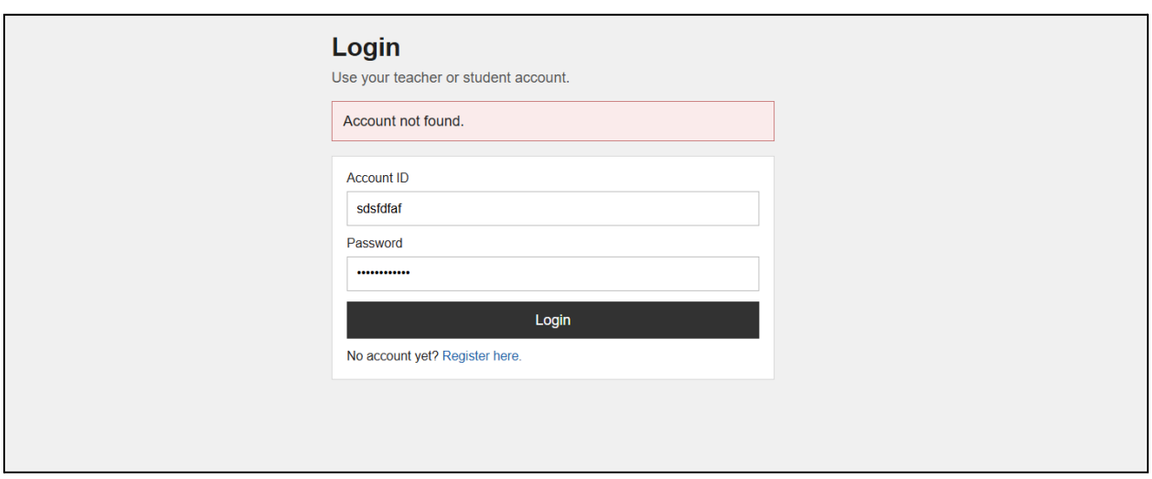


Figure 2.1.3 Logging in with invalid account ID (API returns not found response)

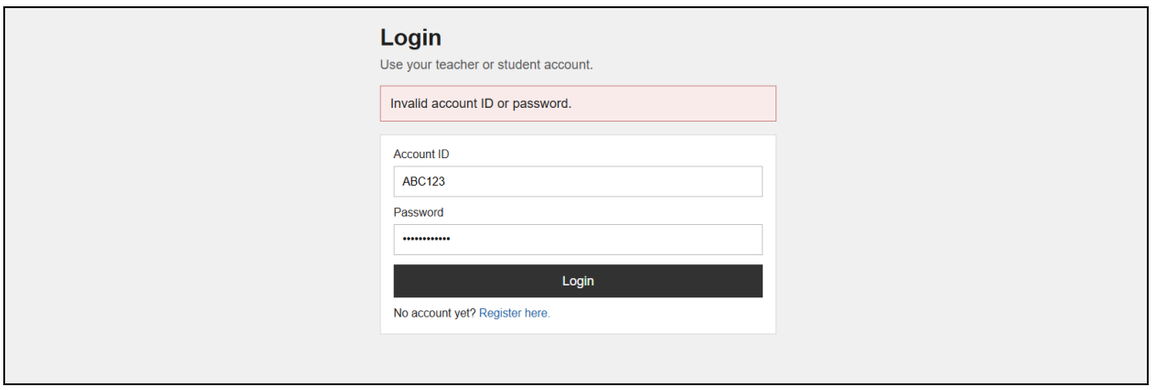


Figure 2.1.4 Logging in with incorrect password (API returns invalid credentials response)

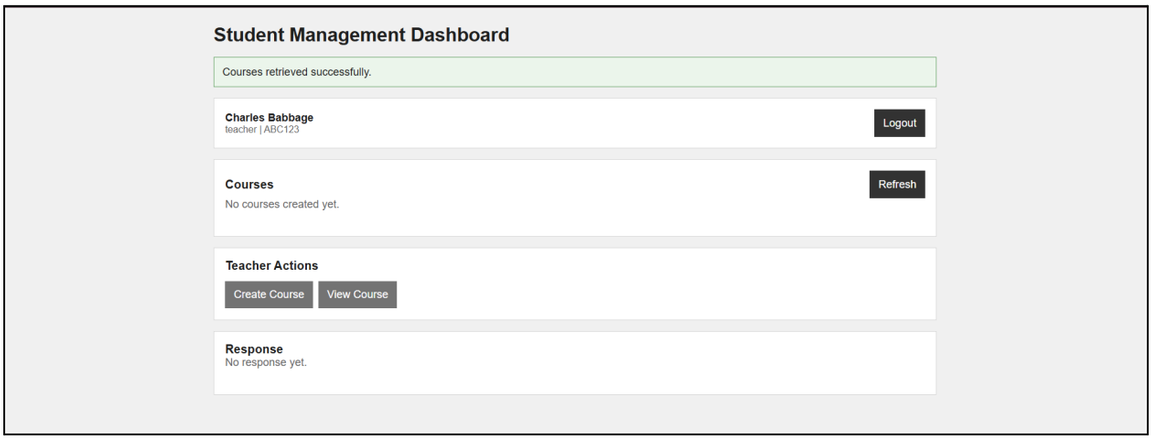


Figure 2.1.5 Main Teacher Dashboard

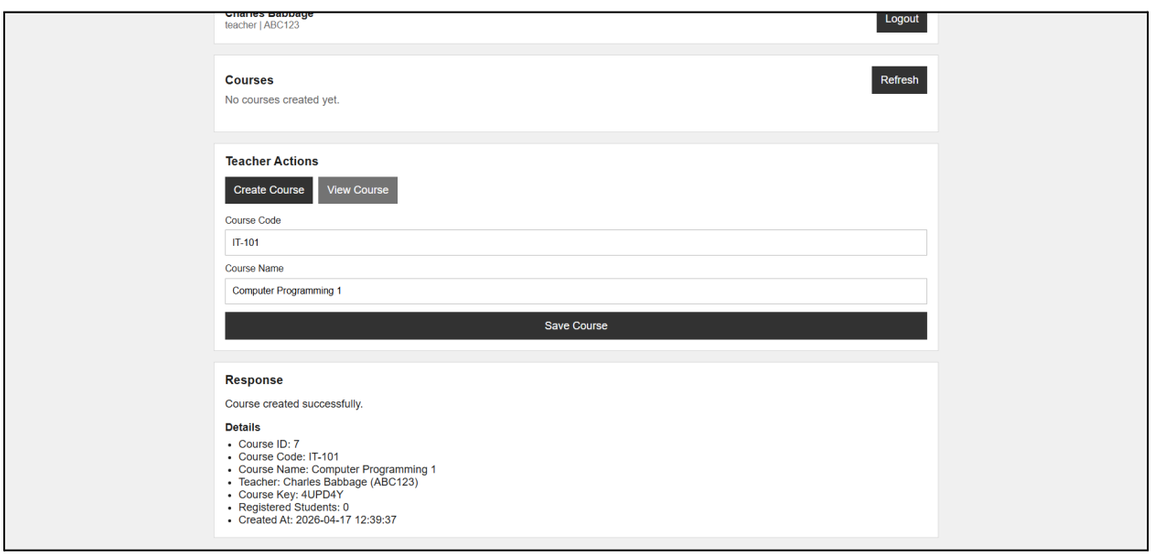


Figure 2.1.6 Creating a Course

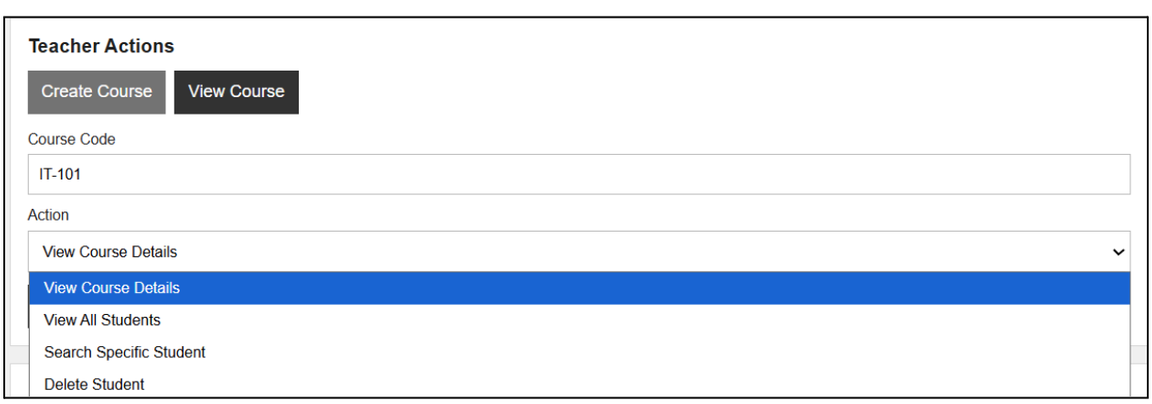


Figure 2.1.7 Options for Viewing a Course

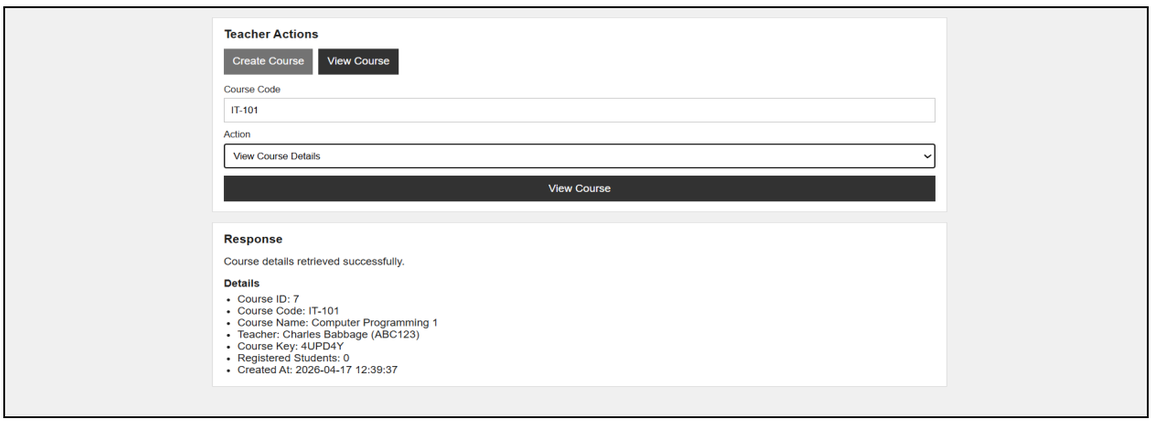


Figure 2.1.8 Viewing a Course Details

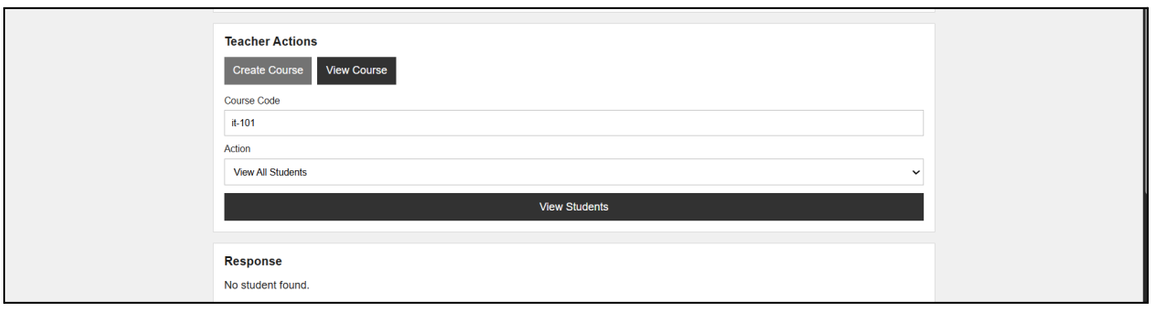


Figure 2.1.8 Viewing All Students (API Returns No Student Found Response)

Figure 2.1.9 Searching for a Student inside the course (API Returns Student Not Found Response)

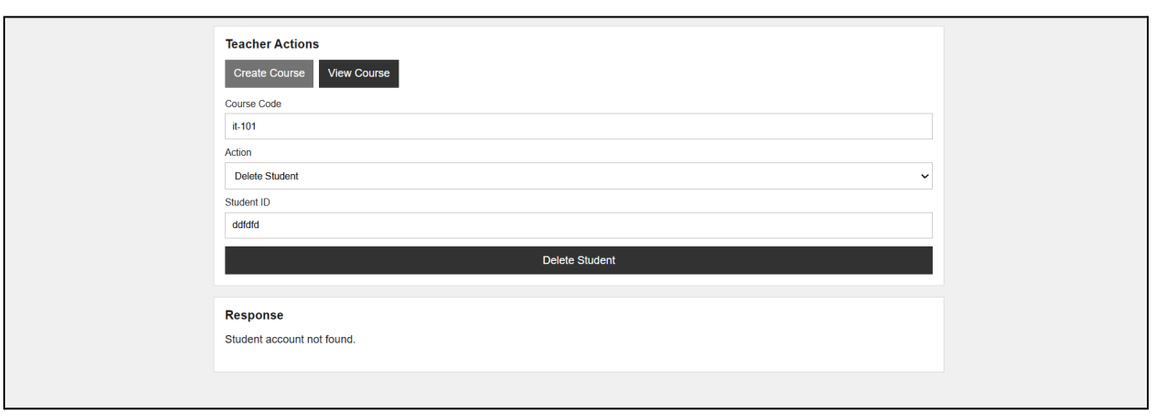


Figure 2.1.10 Trying to Delete a Student (API Returns Student Not Found Response)

2.2 Student-side Functionalities

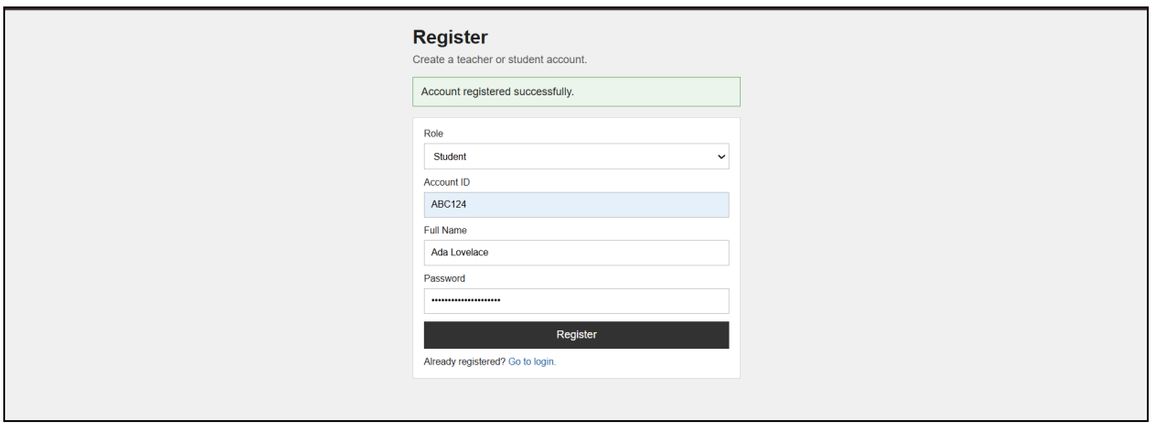


Figure 2.2.1 Registering a Student (same validations as teacher registration)

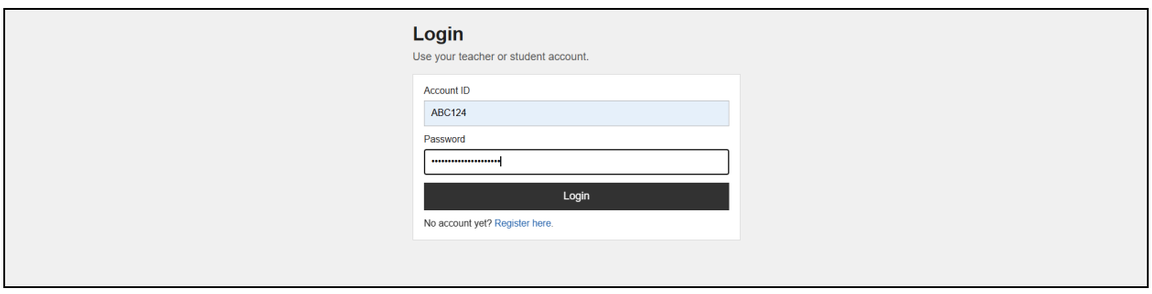


Figure 2.2.2 Logging in as a Student (same validations as teacher login)

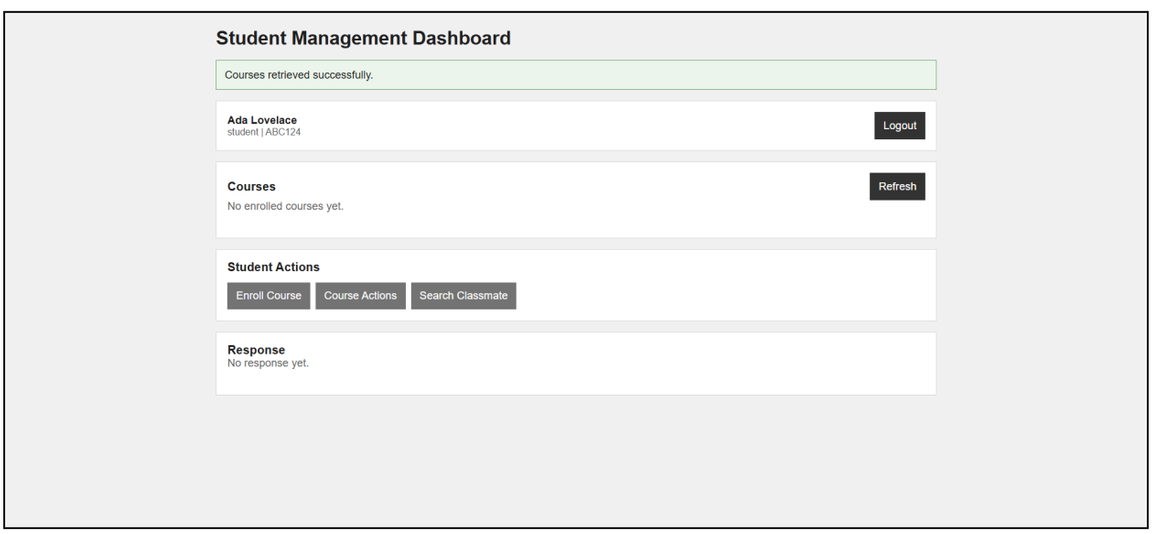


Figure 2.2.3 Main Student Dashboard

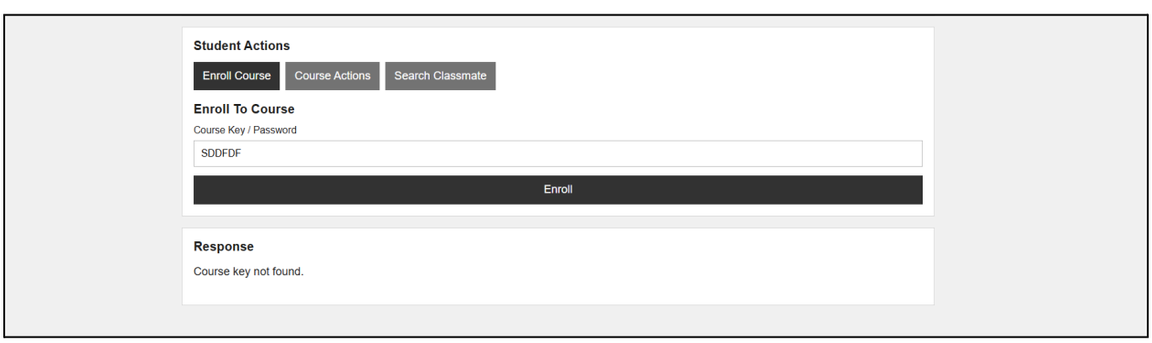


Figure 2.2.4 Invalid Course Key

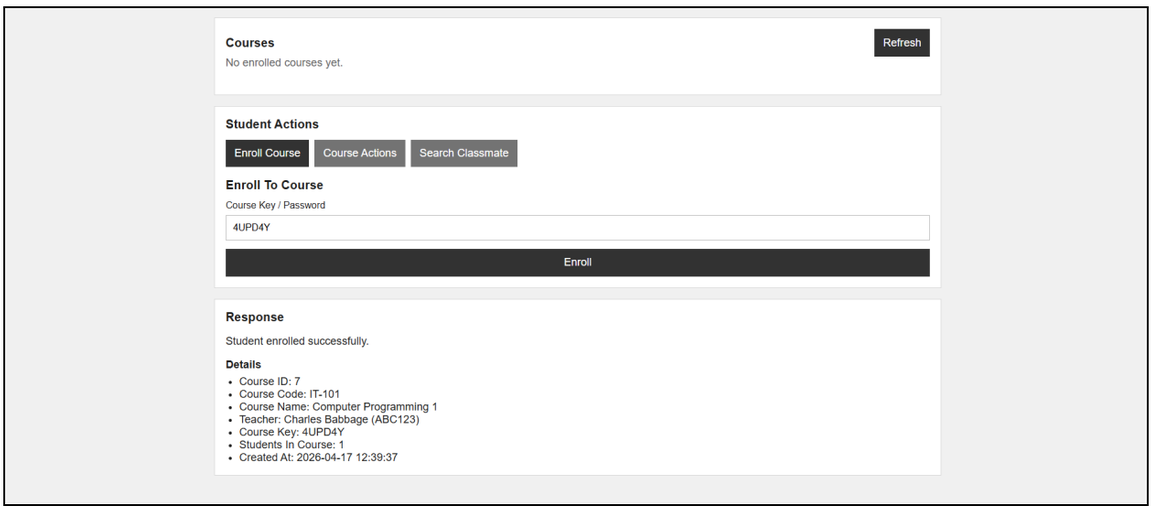


Figure 2.2.5 Valid Enrollment

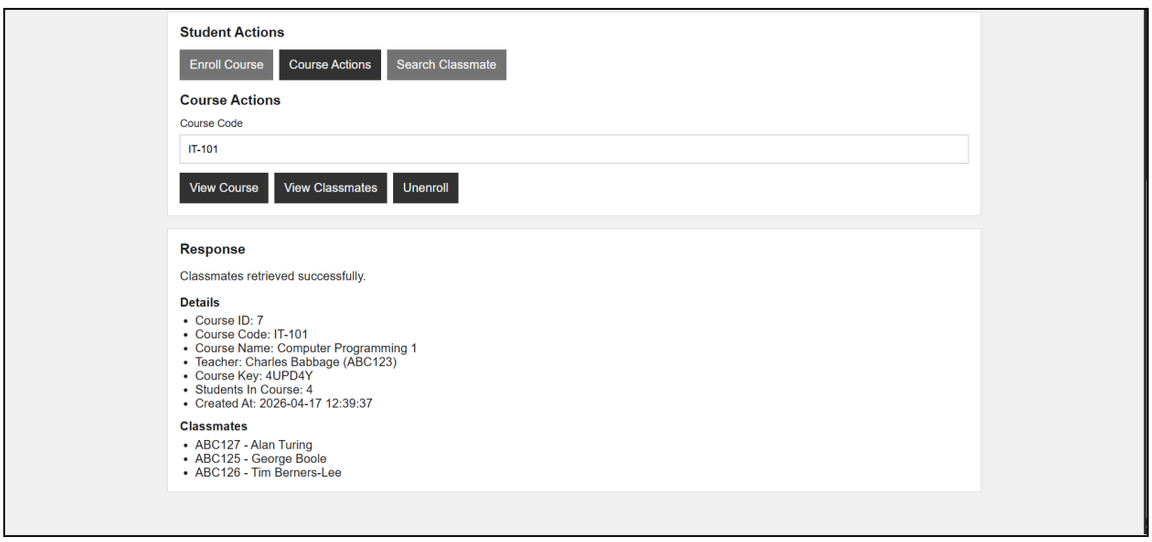


Figure 2.2.6 Viewing Course Classmates (students were added)

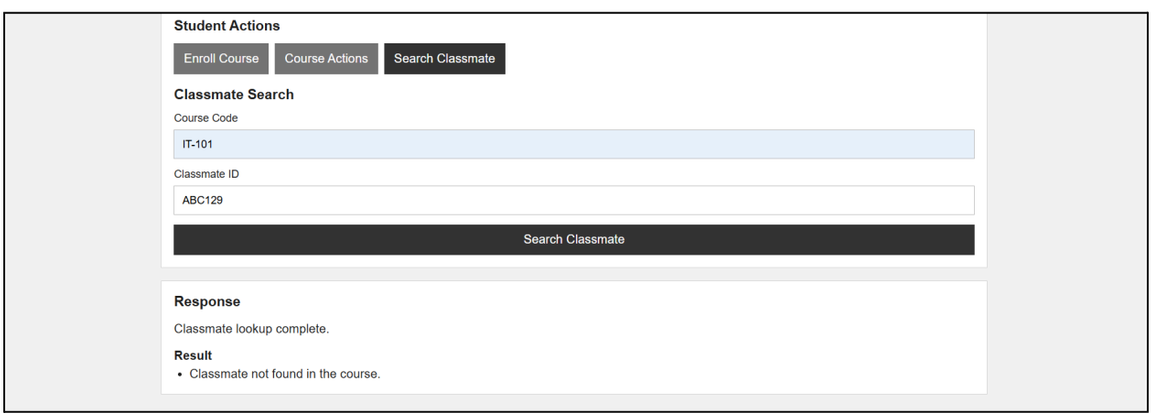
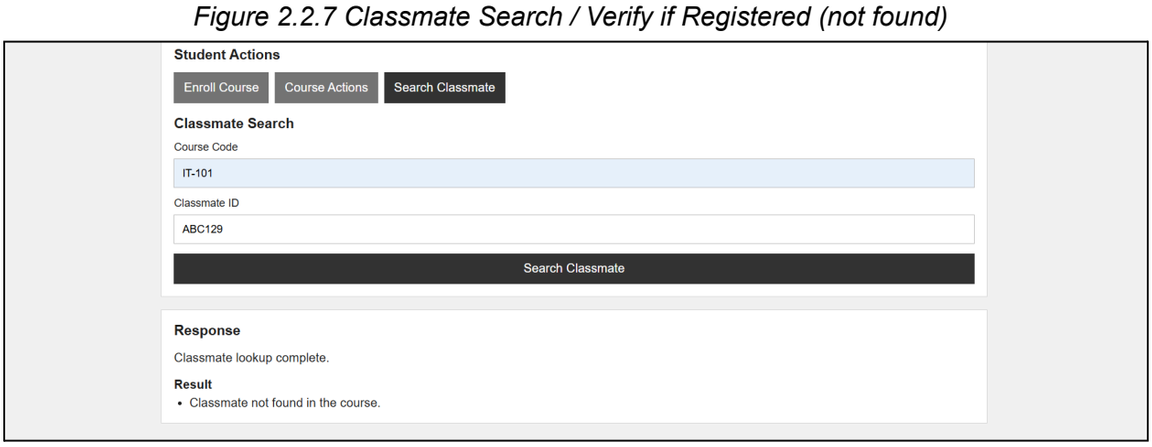


Figure 2.2.7 Classmate Search / Verify if Registered (not found)



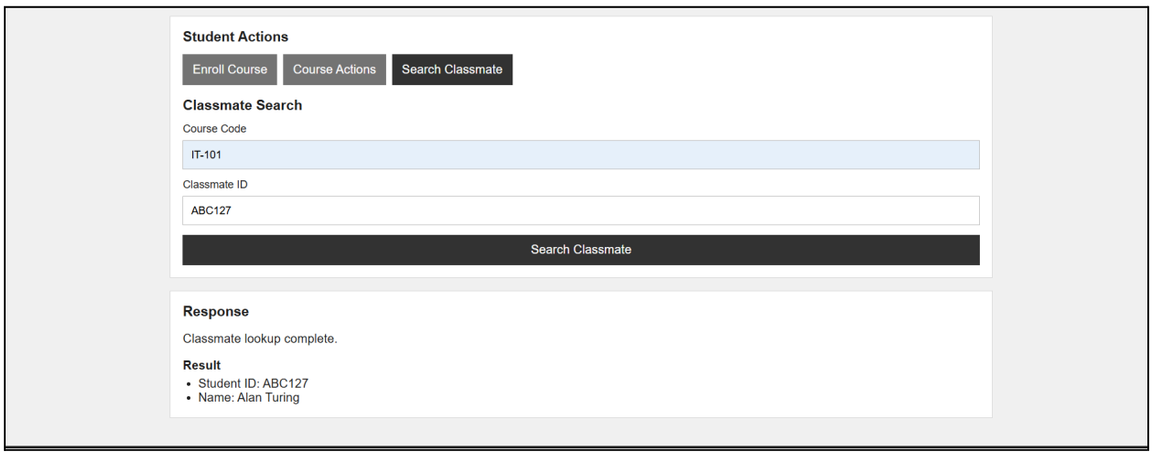


Figure 2.2.8 Classmate Search / Verify if Registered (valid)

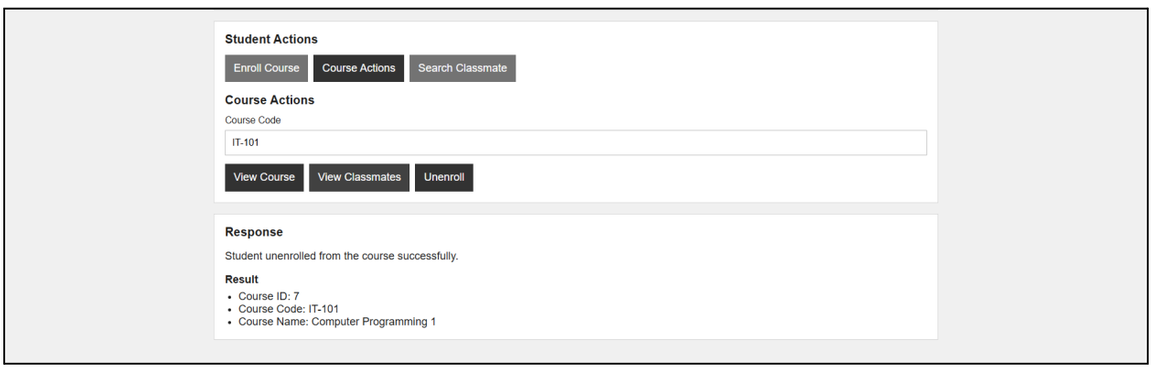


Figure 2.2.9 Student Unenrollment

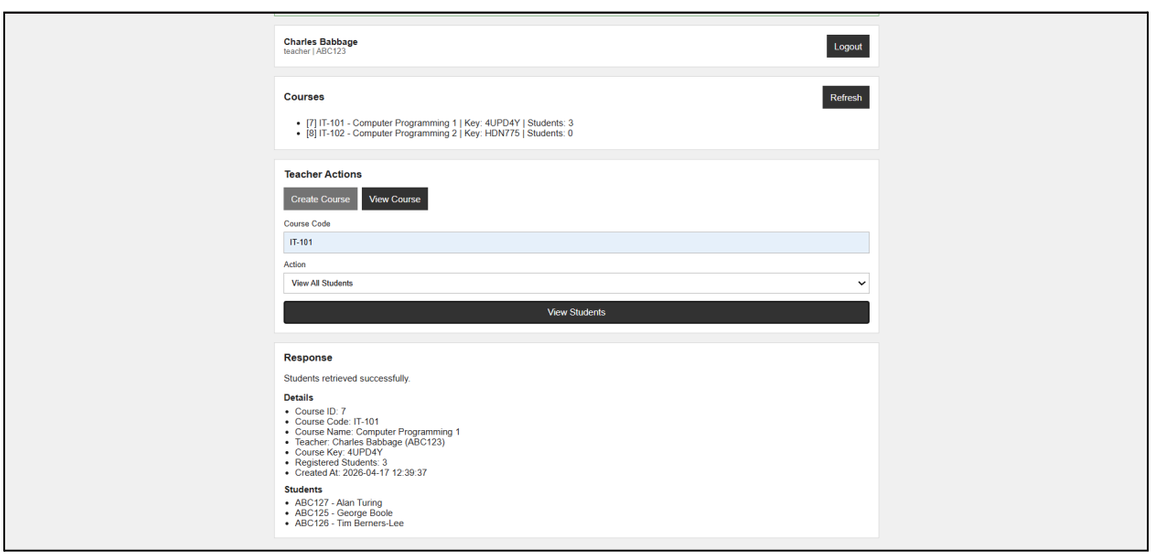


Figure 2.2.10 Teacher Viewing all Students Left Enrolled on Course